A Statistical Framework for Weighted Distribution Aggregation in Heterogeneous Federated Synthetic Data Generation

Final bounded experimental report — 25 October 2026 Owner: Shivansh Gupta, IIT Guwahati BTP **State:** Experimental work complete with explicit scope limits. This is a research artifact for supervisor review, not a publication or proof of a novel algorithm.

Abstract

We tested whether permitted statistical calibration of non-neural client generators improves empirical synthetic-data fidelity over size weighting while preserving a fixed client-population target. Official UCI Adult supplied 32,561 training and 16,281 official-test rows. A corrected projected development matrix comprised 54 matched copula, Chow–Liu and mixed Gaussian-mixture cells, three independent controls, 741 method rows, and five synthetic draws per method/cell. Under $||w-\alpha||_1 \le 0.25$, $w_i \ge 0.5\,\alpha_i$ and a registered 0.005 practical query-max margin, zero of 18 development slices improved practically; mean constrained-minus-size error was +0.002428. Validation selected one projected copula size-weighted procedure, which was frozen before official test. Its 12-cell official confirmation gave overall size query-max 0.022216 and constrained-minus-size −0.000215; zero of six partition/K slices reached the margin. A 12-cell UCI Bank Marketing replication likewise had zero of four practical-gain slices. Secondary utility and conditional findings exposed limits beyond marginal query fidelity. The main model release is nonprivate and, in three measured cells, larger than a comparably serialized raw table. The defensible conclusion is a bounded negative comparison with a specified size-weight fallback, not equivalence, impossibility, end-to-end privacy, or broad population generalization.

1. Estimand and design

Every client holds the same predictor/label schema. Client i fits a non-neural joint generator Q_i from its own training rows and releases that model plus specifically permitted metadata. The target mass is fixed at $\alpha_i = n_i / \sum_j n_j$ from declared client training counts. A method's synthesis weight w_i changes its generator contribution but never redefines α. For a registered query q, the primary real answer is the **alpha-standardized empirical** client reference $\sum_i \alpha_i \, \mathrm{mean}_{D_i^{\mathrm{reference}}} q$; an ordinary pooled held-out answer is saved separately. The synthetic answer is the mean over exactly $m=10{,}000$ generated rows. The primary loss is $\max_q |b_S(q)-b_R(q)|$ over $Q_{\mathrm{primary_v2}}$. Differences are candidate minus tuned projected size, so negative favours the candidate. Adult fnlwgt remains an ordinary predictor, not a survey expansion weight. This is measured dataset-to-dataset discrepancy, not unknown-population TV.

Adult official training duplicates of official test were quarantined before development splitting (25 rows), with remaining full-record duplicate groups kept together. Train/local tune/calibration/validation/development assessment were separately assigned; assignment rules and preprocessing were fit only from permitted early roles, then fixed for held-out roles. No official test row or confirmation seed entered method selection. Public bounds/vocab and structural projection are documented in the release contract. The privileged evaluator could access real held-out rows and candidate per-record scores; deployment-style summary methods received only allowed models, pilots and aggregate releases. F1–F3 predictive scoring used privileged common-real calibration access, which is not equated to model-only access. The main pipeline is nonprivate.

Local model families were independent marginals, integer-bin Chow–Liu, empirical-marginal Gaussian copula and mixed GMM; latent class and KDE explorations are separately versioned. Local six-setting grids and tune criterion are in the immutable freeze and model cards. Exact integer-bin likelihood includes within-bin mass. Chow–Liu's projected joint probability was checked by summing deterministic-projection preimages; copula and rounded/clipped GMM scores remain approximate and are not compared across different base measures. Projection repairs education/education-number on generated Adult rows; structural-zero diagnosis includes capital columns. These corrections were assessment informed and are transparently labelled exploratory development.

The corrected Adult development core covered three generators, age/age+education/label skews, K5/K10 and seeds101/202/303, with five synthesis repeats. The final generator selector used **only validation** primary loss for projected size weighting, equal mean across 18 cells per family and average of five draws within each. Copula scored 0.029975 versus Chow–Liu 0.031455 and mixed GMM 0.051446. The selection rule's 0.001 tie window did not apply. w=alpha, deterministic largest-remainder allocation, tuning grid, workload, conditional support, predictive calibration and confirmation commands were then frozen globally. The freeze SHA-256 is 2420f737b6af732925f9cce3fbb82c8a9a7a30e1988dde5e68900d1404bb3a23.

2. Primary comparisons

The corrected Adult development benchmark has 54 core cells plus three independent controls, 741 method rows and 57 successful finite-pilot LP certificates. Constrained calibration improved some individual cells but **0/18** three-seed generator/partition/K slices reached the registered practical gain. Its overall constrained-minus-size mean was **+0.002427903**. Candidate rules include all eight specified statistical families and uniform/size baselines; formula and access distinctions are in docs/WEIGHT_METHODS.md. Unconstrained apparent gains could carry large client-mass shifts; changing w is not a target change authorized by the estimand. A fixed-pilot LP certificate establishes optimality only for its synthetic matrix, empirical target and constraints.

The Adult confirmation state machine first **scored** the official 16,281-row test at **2026-10-25T05:40:33.072893Z**, after the accepted freeze. Raw test records had earlier been read for source-integrity checks and exact train/test duplicate quarantine, without experiment scoring; docs/TEST_ACCESS_SCOPE_CLARIFICATION.md documents this distinction. All **12** registered copula cells (three partitions × two K × seeds404/505) completed primary and secondary scoring. Each of size, uniform and constrained used five 10,000-row draws from the same per-cell fitted client models. The equal-cell mean size loss was **0.022215720**. Constrained minus size was **−0.000214907**, far short of −0.005; uniform minus size was **+0.041139** overall, notably adverse on label skew. None of the six partition/K two-seed slices passed the gain margin. The table reports/final_build/confirmation_slices.csv contains every slice, including adverse ones. Mean size audit-query max was 0.022552 versus constrained 0.022294, but the audit workload had been viewed during development. Named worst-query counts and all per-query answers are retained. No test-derived threshold or method change was made.

The prospectively frozen official UCI Bank Marketing bank-additional-full.csv external task used 41,188 rows and omitted post-contact duration for pre-call prediction. Its 12 registered cells used two Adult-validation-selected generator families, two hard skews, K5, seeds611/622/633 and five draws/method. Zero of four three-seed slices reached the transferred −0.005 margin; mean constrained-minus-size was **−0.001331749**. It is a different dataset but can contain repeated people without IDs, so it is not a person-independent cohort confirmation. No Bank result reopened Adult selection.

3. Mechanisms, conditional synthesis and prediction

Repeated pilot experiments at four sizes/five seeds and synthetic-size curves at 2,500/10,000/40,000 rows separated pilot/sampling variation from persistent model error. Six-cell paired downstream studies, 12 same-row projection checkpoint contrasts, six release-summary ablations and six fixed-quantity heterogeneity cells examined why error moved. The controlled heterogeneity experiment fixed K5 client training/tune quantity at 1,000/200, compared severity 0.25/0.85 and used a separate equal-client target; it does not isolate every causal feature of assignment. Pure mixtures of overlapping subset mixtures reduce to an effective global mixture; no extra within-subset expressive model is claimed.

Official-test paired probability predictions for logistic and extra-trees were trained and calibrated using separate source-specific roles. Size synthetic logistic averaged **Brier 0.129818 / AUROC 0.856167**, versus privileged real TRTR **0.101987 / 0.904646**; extra-trees similarly favoured TRTR (Brier 0.133218 versus 0.105741). Constrained synthetic logistic Brier was 0.129254, a descriptive small change relative to size. The same held-out people were scored within each cell. Unsupported small-group AUROC is flagged rather than fabricated. Exact projected Chow–Liu joint-implied probabilities, approximate GMM/coupled routes and a separately trained synthetic predictor were compared on matched development people; no exact mixed continuous posterior claim survives.

The deployable conditional default uses the selected teacher with structured public-schema Boolean parsing, a separate 30,000-row support pilot, minimum 300 matches and 1% empirical pilot mass, direct rejection to 1,000 accepted features and a 200,000-proposal cap. It abstains if support or proposal limits fail. In frozen Adult confirmation, three common registered conditions succeeded in all 12 cells with zero accepted-condition violations. Mean within-condition feature-query max was **0.08379** for age25–45, **0.37500** for female and **0.14701** for bachelors+age30–60. The older-doctorate request abstained 12/12. Previously trained per-condition Chow–Liu students remain bounded alternatives; their raw condition violation and within-condition errors were retained. Support is a service heuristic, not a certified probability or fidelity lower bound. Elementary finite-query ratio and student/direct triangle checks are documented in Experiment 11; rare denominators can make bounds vacuous.

4. Published references, privacy and resources

Four maintained Private-PGM/MBI nonprivate fits used the same unnoised aggregate one/two-way measurement access in registered Adult development cells; one small cell improvement and three adverse cells yielded mean PGM-minus-size **+0.002615**. Four adapted FLAIM author-code private fits used epsilon4/delta1e−6 in a separate environment, one private fit per cell and five postprocessing draws; one cell improved, three worsened relative to a **nonprivate** size reference (mean **+0.012894**). Its raw-client interactive private marginal contract is different from BTP's released-model access, so that descriptive utility difference is not an equal-access/privacy ordering. HeteroFedSyn source and protocol were inspected but not numerically replicated because its bundled split/domain and interactive privacy setup differ; no superiority claim is made. Exact commits/patches and scientific contract are in docs/PUBLISHED_REFERENCE_FINDINGS.md.

The full-release ledger classifies models, query/kernel summaries, score evaluations, client identifiers/masses, support decisions, synthetic data and downstream models. The main pipeline's unnoised models/summaries are **nonprivate**. A separate public-support, structural-zero-aware Laplace histogram has a conditional fixed-membership/replacement-adjacency guarantee; releasing epsilon1,4,8 versions on overlapping records incurs basic epsilon13 composition before additional releases. Four overlapping adapted FLAIM epsilon4 fits are separate releases, not a privacy certification of BTP. Fresh private noise uses OS entropy. Attack AUC is merely diagnostic. Three selected copula application-payload comparisons under identical pickle5+gzip6 encoding showed model plus metadata **1.915–2.866 times** raw client-table bytes. Application payload/socket timings are not WAN/TLS traffic or secure deployment measurements. No dollar spending estimate is invented.

5. Evidence, uncertainty and limitations

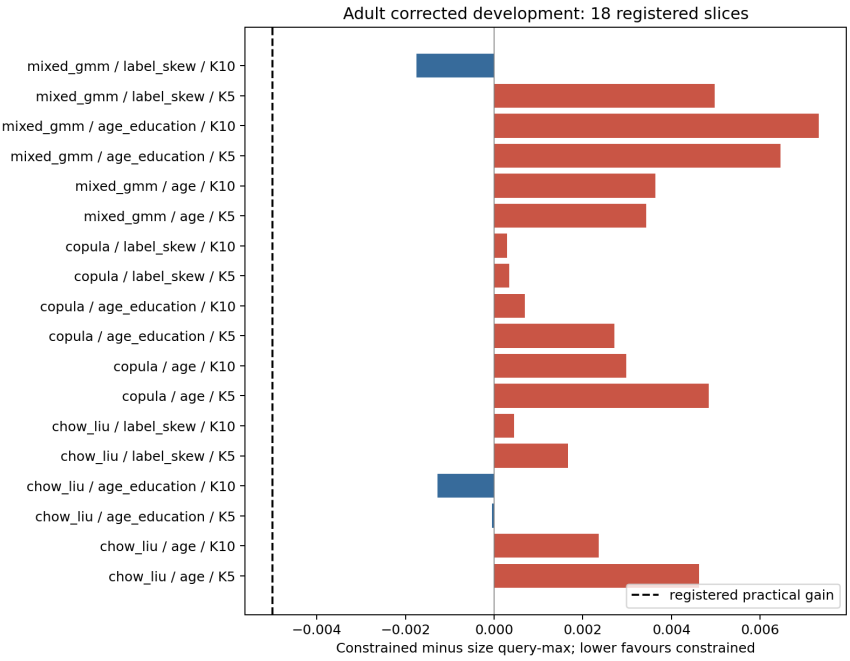
The report builder scripts/build_final_reports.py consumes saved artifacts and emits reports/final_build/summary.json, per-draw/cell/slice/query/utility/conditional CSVs and static scientific figures. ALL_RUNS_INDEX.csv/json separates historical, corrected current, external, reference and confirmation evidence; failed attempts remain in ALL_FAILURES_AND_REPAIRS.md. The clean-room check regenerated a confirmation size draw byte-identically (SHA-256 973e04cbd4e9845031ffc2d30532e6102a0beaa5ab4bfb75e0e0bd9ff498c30e) and reconstructed its maximum named query. The pre-freeze suite had **91 passing tests**. Tests and reproducibility establish software/artifact consistency, not population validity.

Adult test people recur across partition/seed configurations, Bank contacts may recur, and five draws estimate synthetic sampling conditional on fixed models. No p-value treating those draws as independent populations is reported. The main maximum is nonsmooth and limited to 152 registered primary queries; audit queries were viewed during development. Real sampling, assignment, model fit, pilot, draw, optimization and DP-noise uncertainty are distinct. Alpha mass constraints do not by themselves guarantee observable subgroup fidelity. Copula's nominal ordering and approximate scores, residual model misspecification, supported-condition errors and payload inflation limit deployment claims. The empirical thesis is a rigorous bounded negative comparison and size-weight fallback. General method superiority, end-to-end DP, arbitrary-query/condition transfer and publication novelty are not established.

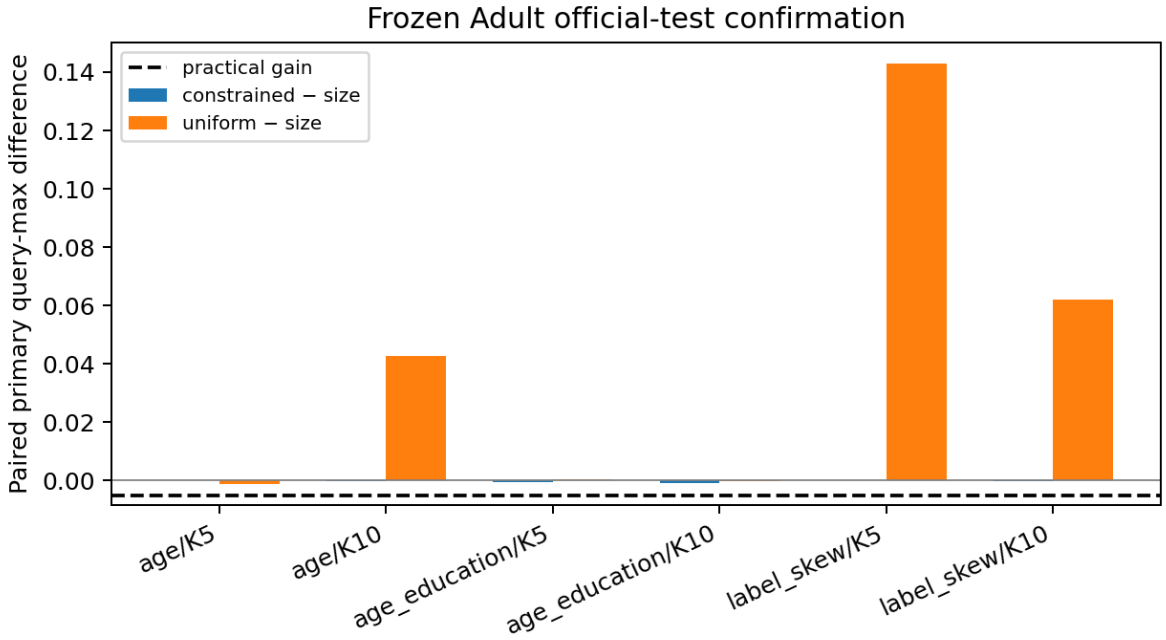
6. Reproduction and handoff

The twelve-package row-by-row closure is EXPERIMENTS_01_TO_12_CLOSURE.md; the exact protocol and version history are in SELECTION_FROZEN.json, EXTERNAL_PROTOCOL_FROZEN.json, docs/ANALYSIS_PROTOCOL_20261024.md and docs/EVIDENCE_INDEX.md. REPRODUCE.md gives executable commands and roles. THEORY_ONLY_BACKLOG.md separates mathematical proof targets from the explicitly outside-scope empirical work in EXPERIMENTAL_BLOCKERS.md. The final archive manifest verifies included bytes; trusted local loading is required for pickle checkpoints. No external publication or repository push occurred.

Adult corrected development slices



Frozen official Adult confirmation



External Bank replication

